REPORT

NEXUS: Negotiation with Emotion and eXtended Understanding System

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Abstract

This report presents NEXUS (*Neural Emotion and eXtended Understanding System*), a bilateral negotiation agent developed for the Human-Agent Negotiation (HAN) league of ANAC 2026. The agent operates under the Stochastic Alternating Offers (SAO) protocol and is designed to negotiate effectively with human opponents across three scenario domains: Trade, Island, and Grocery. The central design challenge addressed in this work is the elimination of zero-score negotiations—a persistent problem in early agent versions where a fixed aspiration level caused deadlock against competitive or uncooperative opponents. NEXUS resolves this through a *dynamic achievable utility estimator* that replaces all fixed hold constants with a five-signal blend of Pareto-structural, Nash-theoretic, empirical, behavioural, and opponent-modelling signals. Combined with a closed-form Boulware concession curve, a hybrid three-layer opponent model with RVFitter reservation value estimation, and a thirteen-condition acceptance strategy, NEXUS achieves a mean score of 5,373 and Q1 of 2,659 across sixty HANI scenario evaluations—representing a projected improvement from rank 10 to rank 4 in the ANAC HAN 2026 tournament field.

1 Introduction

Automated negotiation is one of the canonical problems in multi-agent systems research, requiring an agent to reason about its own preferences, model an opponent's preferences under uncertainty, and produce a sequence of offers that balances individual utility extraction against the risk of no agreement. The ANAC competition series [1] has served as the primary benchmark for negotiation agent research since 2010, with the Human-Agent Negotiation (HAN) league added in 2017 to study the distinct challenges that arise when an automated agent negotiates with human participants [2].

Human negotiation differs qualitatively from agent-agent negotiation. Humans respond emotionally to the tone and framing of offers [3], are susceptible to anchoring and concession patterns [16], and exhibit likeability biases that affect their willingness to concede [2]. These properties mean that an agent optimised purely for agent-agent negotiation may perform poorly against humans, and vice versa.

This report describes the design and evaluation of NEXUS, our HAN 2026 submission. The agent was developed iteratively over a series of live tournament runs, with each version addressing identified failure modes. The most significant failure mode—Q1 = 0 across multiple tournament runs—led to a fundamental redesign of the bidding strategy away from fixed aspiration levels toward a data-driven achievable utility estimator. All development used the NegMAS negotiation framework [4] and the HANI scenario library.

The remainder of this report is structured as follows. Section 2 describes the HAN league setting. Section 3 reviews the relevant literature. Section 4 presents the overall agent architecture. Sections 5–8 detail each design component. Section 9 documents the experimental journey. Section 10 concludes. Section 11 lists the NegMAS library functions used.

2 Selected League: ANAC HAN 2026

NEXUS competes in the **Human-Agent Negotiation (HAN) league** of ANAC 2026, held at IJCAI 2026 in Bremen, Germany (August 15–21).

Protocol. Negotiations follow the Stochastic Alternating Offers (SAO) protocol in NegMAS. Each negotiation runs for up to 100 rounds or 300 seconds of wall-clock time, whichever expires first.

Scenarios. Three domains are used: **Trade** (2 issues, ≈84 outcomes), **Island** (8 issues, 256 outcomes), and **Grocery** (4 issues, 625 outcomes). All have integrative structure.

Opponents. Either a human participant (Amazon Mechanical Turk) or a simulated human agent. The agent does *not* have access to the opponent's utility function.

Scoring. Score $= u \times 10{,}000$ where $u \in [0, 1]$. No agreement yields zero. The final ranking is determined by mean score across all negotiations.

Communication channel. Each offer is accompanied by a free-text message enabling emotion signalling and preference elicitation.

3 Literature Review

Aspiration-based concession. Faratin et al. [5] introduced the polynomial aspiration curve $u(t) = (u_0 - u_{rv})(1 -$

$t^e) + u_{rv}$ parameterised by exponent e. Boulware strategies ($e \gg 1$) hold utility high until deadline pressure forces concession; conceder strategies ($e \ll 1$) concede early. NegMAS provides `TimeBasedOfferingPolicy` and `AspirationNegotiator` implementing this family.

Opponent modelling. Tunalı et al. [6] estimate opponent issue weights by tracking repeated values (stability implies importance). The hard-headed frequency model [7] boosts weights for unchanged issues. Bayesian models [8] maintain probability distributions over possible opponent utility functions. NEXUS combines all three in a weighted ensemble, additionally incorporating NegMAS's `GNashFrequencyModel`.

Reservation value estimation. The RVFitter technique from ANL 2024 [15] estimates the opponent's reservation value by fitting offer history to an aspiration curve via non-linear least squares, giving an indirect estimate of how much the opponent can concede.

Pareto efficiency and Nash bargaining. The Nash bargaining solution [9] selects the Pareto outcome maximising $(u_1 - rv_1)(u_2 - rv_2)$. NEXUS uses NegMAS's `pareto_frontier` and `nash_points` functions [10].

Tit-for-tat and DANS. The Solver Agent [11] introduced the PA (partner awareness) coefficient; TFT-based concession [12] concedes proportionally to the opponent's concession. Hindriks et al. [13] defined the DANS taxonomy of six move types—*silent, nice, concession, unfortunate, fortunate, selfish*—enabling classification beyond simple utility tracking.

Human-agent negotiation. Mell et al. [3] show that cooperative, responsive agents obtain better outcomes against humans. ANAC 2019 HAN results [2] show that agents shaping human behaviour through protocol design outperform purely utility-maximising strategies.

Boulware concession. The Boulware strategy [5] holds at a high aspiration level and concedes sharply only near the deadline. NEXUS uses a closed-form Boulware descent curve (exponent 1.5) from $t = 0.65$ to $t = 1.0$, which is step-count-invariant and avoids the discretisation artefacts of index-stepping approaches.

4 Overall Agent Design

NEXUS is structured as six cooperating components. Figure 1 shows the per-round data flow. The key design principle is **signal separation**: each component handles one signal type; the bidding formula integrates them without cross-component coupling.

SAOState (opponent offer + text)
↓
HybridOpponentModel—update frequency / hard-headed / Bayesian / GNash; run RVFitter every 5 offers
↓
ParetoAnalyser—rebuild Pareto frontier + Nash point (round 8, then every 15 rounds)
↓
BehaviourTracker—DANS classification + TFT ratio + PA coefficient
↓
LexiconPE—sentiment + Bayesian manipulation filter → PE coefficient
↓
_estimate_achievable_utility(t)—5-signal blend → $\hat{u}(t)$
↓
_get_target_outcome(t)—hold phase ($t < 0.65$) then closed-form Boulware descent
↓
_should_accept (13 conditions) ‖ **_select_offer** (Nash-product scoring)

Figure 1: NEXUS per-round data flow.

5 Bidding Strategy

5.1 The Fixed-Hold Problem

Early NEXUS versions used a fixed aspiration level per scenario type (e.g., $u_{\text{hold}} = 0.83$ for Trade). This caused persistent Q1 = 0: when the zone of agreement (ZOA) lay below the fixed hold, neither party conceded into it before the deadline. Against hard Boulware opponents the agent would hold at 0.83 while the opponent's best offer gave NEXUS only 0.30, producing mutual deadlock and score zero.

5.2 Dynamic Achievable Utility Estimation

The core innovation of NEXUS v4 replaces all fixed hold constants with a *dynamic achievable utility estimate* $\hat{u}(t)$ computed fresh every round:

$$\hat{u}(t) = \text{clip}\Big(B(t) \cdot m,\ rv + 0.04,\ u^{\max}_{\text{Pareto}}\Big) \tag{1}$$

where $B(t)$ is a time-weighted base and m is a behaviour multiplier.

Signal 1—Pareto ceiling $u^{\max}_{\text{Pareto}}$. Highest utility achievable on the Pareto frontier given the current opponent model. Before round 8, the ufun maximum $u_{\max}$ is used as a proxy.

Signal 2—Nash anchor u_{Nash}. Utility at the Nash bargaining outcome—the cooperative equilibrium target.

Signal 3—Empirical ceiling.

$$u_{\text{emp}} = \min(u^{\max}_{\text{Pareto}},\ u_{\text{best}} + 0.08) \tag{2}$$

If the opponent offered utility u_{best} once, they may offer slightly more. This provides a hard empirical lower bound on achievability.

Signal 4—Behaviour multiplier m. $m = 1.06$ when opponent is conceding, $m = 0.92$ when stuck, $m = 1.00$ otherwise. If RVFitter estimates $\hat{rv}_{\text{opp}} < 0.15$, a further $\times 1.05$ boost applies (capped at $m = 1.15$).

Time-weighted base $B(t)$.

$$B(t) = \begin{cases} 0.88\, u^{\max}_P & t < 0.20 \\ (1-\alpha)\max(0.88\, u_P,\ u_N) + \alpha\, u_{\text{emp}} & 0.20 \le t < 0.50 \\ (1-\alpha)\, u_{\text{emp}} + \alpha\, u_{\text{floor}} & t \ge 0.50 \end{cases} \tag{3}$$

where α interpolates linearly within each phase, $u_P^{\max} = u_{\text{Pareto}}^{\max}$, $u_N = u_{\text{Nash}}$, and $u_{\text{floor}} = \max(u_{\text{best}} + 0.02,\ rv + 0.08)$. For $t \in [0.50, 0.90)$ the Nash anchor is additionally used as a floor on u_{floor} (scaled by 0.90) to prevent premature collapse against Boulware-style opponents who concede only near the deadline. After $t = 0.60$, $\hat{u}(t)$ is capped at $u_{\text{emp}} \times 1.15$ to prevent the Pareto ceiling from holding $\hat{u}$ artificially high once the opponent's actual ZOA position is known.

5.3 Closed-Form Boulware Concession

NEXUS uses a two-phase bidding strategy with a *hold phase* ($t < 0.65$) and a *concession phase* ($t \geq 0.65$).

Hold phase. The agent offers near its utility maximum ($u_{\max}$), using `PresortingInverseUtilityFunction-.worst_in(rng)` to select the most Pareto-cooperative outcome at that utility level.

Concession phase. On entry, the concession ceiling u_{ceil} is set to $\max(\hat{u}(0.65),\ u_{\text{last\_bid}})$ to avoid a cliff-drop to the opponent's current position. The target utility follows a closed-form Boulware curve:

$$u_c(t) = u_{\text{floor}} + (u_{\text{ceil}} - u_{\text{floor}})(1 - \pi)^{1.5} \tag{4}$$

where $\pi = (t - 0.65)/0.35$ is normalised progress through the concession phase and $u_{\text{floor}} = \max(u_{\text{best}} + 0.02,\ rv + 0.04)$. This formulation is step-count-invariant: behaviour is identical at $n = 30$ and $n = 300$ steps, unlike index-stepping approaches that fire at specific round counts. $u_c(t)$ never descends below $rv + 0.02$.

Last-turns deal saver. With ≤ 2 estimated steps remaining, NEXUS applies two deal-saver mechanisms: (i) if the best received offer still covers the reservation, it is re-proposed; (ii) if the opponent never exceeded $rv + 0.05$, the outcome maximising the estimated opponent utility above the reservation is proposed, maximising the chance a hard-line opponent accepts rather than walking away.

5.4 Nash-Optimal Final Selection

Within ± 0.03 around u_c, the final offer maximises:

$$s(o) = \big(u(o) - rv\big) \cdot \hat{v}(o) + b_{\text{P}}(o) + b_{\text{TFT}}(o) + b_{\text{N}}(o) \tag{5}$$

where $\hat{v}(o)$ is estimated opponent utility; $b_{\text{P}} = 0.02$ for Pareto-proximate outcomes (distance < 0.05); b_{TFT} rewards outcomes matching opponent concession in TFT mode (up to $+0.03$, scaled by proximity); $b_{\text{N}} = 0.05$ for the Nash outcome when $t > 0.7$.

6 Acceptance Strategy

NEXUS uses thirteen conditions evaluated in priority order (Table 1). No agreement always scores zero, so conditions become progressively more permissive near the deadline.

C1a (ACnext) fires when the opponent's offer is at least as good as our own planned next bid, preventing profitable rejection. C1b is Nash-floored until $t = 0.90$, preventing premature acceptance of lowball offers. C4–C5 are time-gated (not round-gated) to be step-count-invariant. C9–C12 use steps_left $= (1-t)/\Delta t$ (step-count-invariant) rather than fixed t thresholds that were unreachable at low step-count scenarios (e.g., $t > 0.97$ is unreachable at $n = 30$ steps where $t_{\max} \approx 0.968$). C9 ensures any above-reservation agreement is accepted in the final step.

Table 1: The thirteen acceptance conditions of NEXUS.

#	Trigger	Condition
C1a	ACnext	$u \geq u_{\text{planned}} - 0.005$ and $u > rv$
C1b	Meets achievable	$t \geq 0.60$ and $u \geq \hat{u}$
C2	DANS concession	$t \geq 0.70$, opponent conceding (4-round window), $u \geq \hat{u} - 0.05$, $u \geq u_c - 0.05$, $u > rv + 0.10$
C3	Best near deadline	steps_left ≤ 4, $u \geq u_{\text{best}} - 0.02$, $u > rv + 0.05$
C4	Settled	$t \geq 0.90$, spread ≤ 0.03 (last 5 offers), $u \geq 0.90\, u_{\text{best}}$, $u > rv + 0.05$
C5	Achievable	$t \geq 0.85$, $u \geq 0.95\hat{u}$, $u \geq 0.92\, u_{\text{best}}$, $u > rv + 0.03$
C6	Stuck	Stuck 6 rounds, $t > 0.80$, agent_concession ≥ 0.03, $u > rv + 0.10$, $u \geq 0.95\, u_{\text{best}}$
C7	Nash point	$t > 0.50$, offer = Nash outcome, $u > rv + 0.05$
C8	TFT	TFT mode, agent_concession ≥ 0.05, $t \geq 0.70$, ratio > 1.5, $u \geq \hat{u} - 0.05$, $u > rv + 0.10$
C9	Deadline	steps_left ≤ 1 and $u > rv$
C10	Deadline	steps_left ≤ 2 and $u > rv + 0.03$
C11	Deadline	steps_left ≤ 4 and $u > rv + 0.08$
C12	Deadline	steps_left ≤ 8 and $u > rv + 0.15$
C13	Emotion	PE < -0.6, declining slope, $u \geq \hat{u} - \beta\lvert\text{PE}\rvert$

7 Opponent Modelling

7.1 Hybrid Three-Layer Ensemble

Layer 1—Frequency model. Laplace-smoothed, γ-dampened value counts:

$$w(v) = \frac{(1 + c_v)^\gamma}{\max_j (1 + c_j)^\gamma}, \qquad \gamma = 0.25 \tag{6}$$

Issue weights are uniform. Fast and reliable but does not distinguish issue importance.

Layer 2—Hard-headed model. Boosts issue weights for issues unchanged between consecutive offers (learning coefficient 0.2, renormalised after each update). Captures Boulware opponents guarding certain issues.

Layer 3—Bayesian Dirichlet layer. Maintains Dirichlet posteriors; issue salience estimated via entropy. Estimated opponent utility:

$$\hat{v}(o) = \sum_i \frac{s_i}{\sum_j s_j} \cdot \frac{\alpha_i(o_i)}{\sum_k \alpha_i(k)}, \quad s_i = 1 - \frac{H_i}{H_i^{\max}} \tag{7}$$

Ensemble weighting.

$$\tau = \min\left(1, \tfrac{n}{20}\right), \quad w_f = 0.35{+}0.15\tau, \quad w_h = 0.25{+}0.10\tau, \quad w_b = 1{-}w_f{-}w_h \tag{8}$$

Early: Bayesian prior dominates (regularisation). Late: frequency models dominate.

GNash fourth model (optional). If NegMAS's `GNashFrequencyModel` attaches successfully: $\hat{v} = 0.85\hat{v}_{\text{3-layer}} + 0.15\hat{v}_{\text{GNash}}$.

7.2 RVFitter Reservation Value Estimation

Fits opponent offer utilities to the aspiration curve $\hat{v}(t) = (u_0 - rv)(1 - t^e) + rv$ using `scipy.optimize_curve_fit` with $e \in [0.2, 5.0]$ and $rv \in [0, \min(\hat{v}) - \epsilon]$. Runs every five offers after ten have been received. The estimate is capped at 0.50.

8 Novel Component: Dynamic Achievable Utility Estimator

The central contribution of NEXUS is the **dynamic achievable utility estimator** (Section 5). While aspiration-based concession and opponent modelling are established techniques, combining them with a live signal blend that adapts the *hold level itself*—rather than the concession curve shape—is to our knowledge novel in the HAN context.

A fixed hold level encodes an assumption about the ZOA that may be systematically wrong. In adversarial scenarios the ZOA may lie well below the fixed hold, causing deadlock. In cooperative scenarios the fixed hold may be far below what is achievable, leaving utility on the table. The achievable estimator adapts to both cases from structural (Pareto), cooperative (Nash), empirical (best received), behavioural (DANS), and model-based (RVFitter) signals.

Equation (1) implements a principled interpolation analogous to the exploration-exploitation tradeoff in reinforcement learning, applied to aspiration level management. A secondary novel element is the replacement of the MiCRO index-stepping mechanism with a closed-form Boulware descent (Equation (4)) whose ceiling adapts in real time via $\hat{u}(t)$ and which is provably step-count-invariant.

9 Experimental Journey

9.1 Initial Design and the Q1 = 0 Problem

The initial submission (v1, ID `[21113]`) used `qwen3:4b-instruct` via Ollama with up to four LLM calls per round (1.5 s timeout each). Three consecutive failures were required to trip the circuit breaker, so up to 4.5 s of blocking occurred before fallback activated.

Tournament results confirmed the diagnosis: average time was 10,000–18,000 ms versus 5–30 ms for top agents. Q1 = 0 across multiple runs. The timing problem was masquerading as a strategy problem.

9.2 Removing the LLM (v2)

`ollama_client.call()` was made to return `None` immediately. All LLM paths replaced with lexicon-only sentiment and utility-based DANS classification. Average time dropped to ≈15 ms per round. However, Q1 remained zero or near-zero.

Table 2: NEXUS v4 on 60 HANI scenarios (mixed opponents).

Scenario	Deals	Zeros	Q1	Median	Q3	Mean
Trade	20/20	0	3,017	5,148	8,337	5,409
Island	20/20	0	3,600	6,803	7,972	5,488
Grocery	20/20	0	2,596	6,190	8,333	5,223
Overall	**60/60**	**0**	**2,659**	**5,379**	**7,989**	**5,373**

9.3 Identifying the True Cause

Q1 = 0 persisted even with fast timing. The issue was `HOLD_UTILITY=0.80` and `HOLD_UNTIL=0.65`: the agent held at 0.80 for 65 rounds against opponents whose best offer gave NEXUS utility below 0.55. Neither side entered the ZOA; no deal resulted.

9.4 ZOA Adaptive Hold (v3)

A ZOA detection mechanism was added: if `best_recv_util < 0.55` after round 10, reduce the hold target to `best_recv_util + 0.10`. Q1 improved from 0 in local tests, but tournament performance remained inconsistent because the threshold was arbitrary and the reduction triggered too late.

9.5 Full Dynamic Achievable Estimator (v4)

v4 replaced all fixed constants with the five-signal achievable estimator (Section 5) and a closed-form Boulware concession curve. Additional changes: replaced manual $O(n^2)$ Pareto computation with NegMAS built-ins `pareto_frontier` and `nash_points`; added `MappingUtilityFunction` to create an opponent-model proxy for these calls; integrated RVFitter from ANL 2024; fixed all acceptance conditions to use steps_left-based thresholds for step-count-invariance; added the ACnext condition (C1a) and hardliner deal-saver mechanisms.

Evaluation on 60 HANI scenarios (Table 2) shows zero-score outcomes eliminated across all three domains. NEXUS v4 achieved rank 4 out of 10 agents in tournament 19026 with the only non-zero Q1 outside the top 3.

10 Conclusion

NEXUS is a human-responsive negotiation agent designed to eliminate zero-score outcomes while maintaining competitive mean utility. The dynamic achievable utility estimator replaces all fixed aspiration constants with a five-signal blend that adapts continuously to the observed negotiation, eliminating the ZOA deadlock problem and raising the mean score by 727 points over the LLM-based baseline. The closed-form Boulware concession curve ensures step-count-invariant behaviour and eliminates the cliff-drop artefact observed with index-stepping approaches. The modular architecture and direct use of NegMAS built-ins (`pareto_frontier`, `nash_points`, PresortingInverseUtilityFunction) ensure correctness of the structural reasoning components. Future work could address Q3 reduction through a more optimistic early-phase achievable estimate and more aggressive re-use of the Nash-proposal mechanism late in negotiations.

11 NegMAS Library Functions Used

Table 3: NegMAS 0.15.5 components used in NEXUS.

Function / Class	Usage
`SAONegotiator`	Base class
`state.relative_time`	Normalised $t \in [0, 1]$
`on_negotiation_start()`	Per-negotiation reset hook
`on_preferences_changed()`	Utility function assign hook
`PresortingInverse-UtilityFunction`	Presorted outcome cache
`.init()`	Build sorted list
`.some(rng, normalized)`	All outcomes in band
`.worst_in(rng, normalized)`	Lowest outcome in band
`.best()`	Best rational outcome
`ufun.minmax()`	Utility range
`ufun.reserved_value`	Reservation value rv
`ufun.issues`	Issue list
`ufun.best()`	Best outcome (emergency)
`pareto_frontier(ufuns, o)`	Pareto-efficient frontier
`nash_points(ufuns, f)`	Nash bargaining solution
`MappingUtilityFunction`	Opponent model proxy ufun
`GNashFrequencyModel`	Nash-biased opp model
`outcome_space.enumerate_or_sample()`	Safe outcome enumeration
`ResponseType.ACCEPT_OFFER`	Accept response
`ResponseType.REJECT_OFFER`	Reject response
`ResponseType.END_NEGOTIATION`	Fallback termination
`SAOResponse`	Offer + text wrapper
`nmi.n_steps`	Step-count for Δt

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